

How to compare ranked lists

 stackoverflow.com/questions/13574406/how-to-compare-ranked-lists

November 26, 2012

Asked

Viewed 29k times

21

I have two lists of ranked items. Each item has an rank and an associated score. The score has decided the rank. The two lists can contains (and usually do) different items, that is their intersection can be empty. I need measures to compare such rankings. Are there well-known algorithms (in literature or real-world systems) to do so ? The measure of distance should take into account the scores as well as the ranks of the items.

36

This question has never been answered before, but I still think it's important to a lot of people out there:

Your two requirements, i.e. **non-conjointness of lists** and **importance of ranks** are not met by common correlation tests. In addition to that most of them (Kendall-Tau for example) do not take the order into account:

```
>>> from scipy.stats import kendalltau
>>> kendalltau([1,2,3,4,5], [2,1,3,4,5])
KendalltauResult(correlation=0.79999999999999982, value=0.050043527347496564)
>>> kendalltau([1,2,3,4,5], [1,2,3,5,4])
KendalltauResult(correlation=0.79999999999999982, value=0.050043527347496564)
```

The 1st comparison should yield a significantly smaller value than the 2nd one, because the head of the list is more important than the tail (**2nd requirement**).

In addition to that one can see that both lists need to be the same size and have the same kind of elements (**1st requirement**)

Possible solution:

The measure that satisfies all your needs is called [Rank Biased Overlap](#). It's a generalization of the so called average based overlap, which is wonderfully illustrated in this [blog](#). The same guy also put out an [implementation](#) of RBO.

Update Jan 2018:

Another implementation of [RBO](#) for python 3.5.2

answered Mar 24, 2016 at 10:16



Thanks,, this worked well for me. I have fixed list of stocks [50] and wanted to know how much they were changing day to day with the leading stocks by score getting more weight. Cheers,



Really nice article!

8

Maybe not solving the issue completely, but definitely worth taking a look at [Kendall's weighted tau](#).

It provides a better way of calculating similarity between ranked lists when order matters as i it allows arbitrary weighting based on rank order.

For example, one may be more interested in upweighting similarity in the top 20 items of the list rather than uniformly.

Also has a nice implementation in [scipy](#).

answered Mar 28, 2018 at 14:40



[alex_sp](#)

11311 silver badge77 bronze badges

Comments

1

There are many measures to compare top k (ranked) lists. Some very trivial to compute making several simplifying assumptions, others not so trivial but are more rigorous in their

evaluation of rank similarity between lists. A recent paper I came across that deals with this problem in a statistically meaningful way, using concepts from information theory and data compression: <http://arxiv.org/abs/1310.0110>

answered Jan 28, 2015 at 20:35

user4504777

Comments

Your Answer

Sign up or [log in](#)

Sign up using Google

Sign up using Email and Password

Post as a guest

Name

Email

Required, but never shown

Start asking to get answers

Find the answer to your question by asking.

[Ask question](#)